# Part III

19 January 2024 20:50

**1.** Which of these relations on $\{0, 1, 2, 3\}$ are equivalence relations? Determine the properties of an equivalence relation that the others lack.

*a*) $\{(0, 0), (1, 1), (2, 2), (3, 3)\}$

*b*) $\{(0, 0), (0, 2), (2, 0), (2, 2), (2, 3), (3, 2), (3, 3)\}$

*c*) $\{(0, 0), (1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$

*d*) $\{(0, 0), (1, 1), (1, 3), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$

*e*) $\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$

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Sol: An equivalence relation is reflexive, symmetric and transitive. So, we check this for each relation.

a) construct a matrix rep,

$$I = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \rightarrow$$ diagonal, symmetric

furthermore, $I^{\odot n} = I$ Thus, transitive

Thus, R is an equivalence relation.

• Notice also that R is antisymmetric.

b)
$$\begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{array}$$

not diagonal ones

$\Rightarrow$ R is not reflexive

$\Rightarrow$ R is not an equivalence relation.

c)
$$\begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{array}$$

diagonal $\rightarrow$ reflexive

symmetric $\rightarrow$ symmetric relation

$$M \odot M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \subseteq M \Rightarrow M^{(n)} = M$$

Thus, transitive.

This is an equivalence relation.

d) $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$ diagonal → reflective
symmetric → symmetric action

$M \odot M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & (1) & 1 \\ & & \vdots & \end{pmatrix}$ not transitive since $(2,3), (3,1) \in R$ but $(2,1) \notin R$.

e) $\begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ diagul → symmetric
not symmetric → not a symmetric whit
⇒ not an equivalence relation.

**3.** Which of these relations on the set of all functions from $\mathbf{Z}$ to $\mathbf{Z}$ are equivalence relations? Determine the properties of an equivalence relation that the others lack.

*a*) $\{(f, g) \mid f(1) = g(1)\}$

*b*) $\{(f, g) \mid f(0) = g(0)$ *or* $f(1) = g(1)\}$

*c*) $\{(f, g) \mid f(x) - g(x) = 1$ *for all* $x \in \mathbf{Z}\}$

*d*) $\{(f, g) \mid$ *for some* $c \in \mathbf{Z}$, *for all* $x \in \mathbf{Z}$, $f(x) - g(x) = c\}$

*e*) $\{(f, g) \mid f(0) = g(1)$ *and* $f(1) = g(0)\}$

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a) we need to show that R is reflexive, symmetric and transitive

- reflexive : by definition, $f(1) = f(1)$.
- symmetric : let $f(1) = g(1)$ then $g(1) = f(1) \Rightarrow g \sim f$.
- transitive : let $f(1) = g(1)$ and $g(1) = h(1) \Rightarrow f(1) = h(1)$
  Thus, $f \sim h$ thus, transitive.

This is an equivalence relation.

b) reflexitivity: $f(1) = f(1) \wedge f(0) = f(0)$ Thus, reflexive.
Symmetric: if $f \sim g$ then $f(1) = g(1) \vee f(0) = g(0)$
Then, $g(1) = f(1) \vee g(0) = f(0) \Rightarrow g \sim f$ Thus, symmetric.
Transitivity: if $f \sim g \wedge g \sim h$, then $f(1) = g(1) \vee f(0) = g(0)$
also, $g(1) = h(1) \vee g(0) = h(0)$.

Suppose it is the case that $f(1) = g(1)$, $f(0) \neq g(0)$
and $g(0) = h(0)$, $g(1) \neq h(1)$
Thus, $f(1) \neq h(1)$ and $f(0) \neq h(0)$. Thus, this relation is not transitive

This is NOT an equivalance relation.

c) reflexitivity: $\forall x \in \mathbb{Z}$ $f(x) - f(x) = 0 \neq 1$ so not reflexive -

This is not an equivalance relation.

d) $\exists c \in \mathbb{Z}$, $\forall x \in \mathbb{Z}$ $f(x) - g(x) = c$

reflexitivity: $f(x) - f(x) = 0$, so $C = 0$ $\forall x$.
symmetric: suppose $f \sim g \Rightarrow \forall x$, $f(x) - g(x) = c \in \mathbb{Z}$
$\Rightarrow \forall x\, g(x) - f(x) = -c$, since $(-c) \in \mathbb{Z} \Rightarrow g \sim f$
$\Rightarrow$ The relation is symmetric
Transitivity: suppose $\forall x$ $f(x) - g(x) = c$, $\forall x$ $g(x) - h(x) = m$
$\Rightarrow \forall x$ $g(x) = h(x) + m \Rightarrow \forall x$ $f(x) - h(x) - m = c$
$\Rightarrow \forall x$ $f(x) - h(x) = m + c$, since $m, c \in \mathbb{Z}$
$\Rightarrow m + c \in \mathbb{Z}$. Thus, R is transitive

This is an equivalance relation.

e) $\{f, g \mid f(0) = g(1) \wedge f(1) = g(0)\}$

reflexitivity: choose $f$ such that $f(0) = f(1)$
$\Rightarrow f(1) = f(0)$
symmetric: suppose $f \sim g \Rightarrow f(0) = g(1) \wedge f(1) = g(0)$
$\Rightarrow g(1) = f(0) \wedge g(0) = f(1) \Rightarrow g \sim f$

transitivity: suppose $f \sim g$ and $g \sim h$

$\Rightarrow f(0) = g(1) \wedge f(1) = g(0)$

$g(0) = h(1) \wedge g(1) = h(0)$

$\Rightarrow f(0) = h(0) \wedge f(1) = h(1)$ but $f(0)$ may or may not equal to $h(1)$. Thus, not transitive

This is not an equivalence relation.

**9.** Suppose that $A$ is a nonempty set, and $f$ is a function that has $A$ as its domain. Let $R$ be the relation on $A$ consisting of all ordered pairs $(x, y)$ such that $f(x) = f(y)$.

a) Show that $R$ is an equivalence relation on $A$.

b) What are the equivalence classes of $R$?

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a) $\{(x,y) \mid f(x) = f(y)\}$

reflexitivity: $(x,x) \in R$ since $f(x) = f(x)$ $\forall x \in A$.

Transitivity: suppose $x \sim y$ and $y \sim z \Rightarrow f(x) = f(y)$ and $f(y) = f(z) \Rightarrow f(x) = f(z) \Rightarrow x \sim z$.

symmetric: suppose $x \sim y \Rightarrow f(x) = f(y) \Rightarrow f(y) = f(x)$ $\Rightarrow x \sim y$.

b) suppose $a \in A$. by definition, equivalence classes of $a$, are all elements in $A$ such that $f(a) = f(b)$. Thus, $(a,b) \in R$.

$\Rightarrow [a]_R = \{ b \in A \mid f(a) = f(b), \text{ for some } a \in A \}$.

**15.** Let $R$ be the relation on the set of ordered pairs of positive integers such that $((a, b), (c, d)) \in R$ if and only if $a + d = b + c$. Show that $R$ is an equivalence relation.

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reflexive: $(a,b)$ , $(a,b)$

reflexive : $(a,b)$ , $(a,b)$

since $a+b = b+a \Rightarrow (a,b) \sim (a,b)$

Thus, R is reflexive

symmetric : suppose $(a,b) \sim (c,d)$

$\Rightarrow a+d = b+c \Rightarrow c+b = d+a$

$\Rightarrow (c,d) \sim (a,b)$ Thus, the relation is symmetric

Transitive : suppose $(a,b) \sim (c,d)$

$(c,d) \sim (e,f)$

$\Rightarrow a+d = b+c$

$c+f = d+e$

$\Rightarrow a-b = c-d$ & $c-d = e-f$

$\Rightarrow a-b = e-f \Rightarrow a+f = b+e$

$\Rightarrow (a,b) \sim (e,f)$ #

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**16.** Let $R$ be the relation on the set of ordered pairs of positive integers such that $((a, b), (c, d)) \in R$ if and only if $ad = bc$. Show that $R$ is an equivalence relation.

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$[(a,b),(c,d)] \in R$ iff $ad = bc$

reflexive : $(a,b)$ since $ab = ba$

$\Rightarrow [(a,b),(a,b)] \in R$ Thus, R is transitive.

symmetric : let $[(a,b),(c,d)] \in R \Rightarrow ad = bc$

$\Rightarrow \frac{a}{b} = \frac{c}{d} \Rightarrow cb = da \Rightarrow [(c,d),(a,b)]$

Transitivity : let $[(a,b),(c,d)]$ & $[(c,d),(e,f)] \in R$

$\Rightarrow ad = bc$ & $cf = de \Rightarrow \frac{a}{b} = \frac{c}{d}$ & $\frac{c}{d} = \frac{e}{f}$

$\Rightarrow \quad ad = bc \;\&\; cf = de \Rightarrow \frac{a}{b} = \frac{c}{d} \;\&\; \frac{c}{d} = \frac{e}{f}$

$\Rightarrow \quad \frac{a}{b} = \frac{e}{f} \Rightarrow af = be \Rightarrow [(a,b),(e,f)] \in R$

**17. (Requires calculus)**

a) Show that the relation $R$ on the set of all differentiable functions from $\mathbb{R}$ to $\mathbb{R}$ consisting of all pairs $(f, g)$ such that $f'(x) = g'(x)$ for all real numbers $x$ is an equivalence relation.

b) Which functions are in the same equivalence class as the function $f(x) = x^2$?

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a) $f : \mathbb{R} \longrightarrow \mathbb{R}$ $\quad f^{(1)}$ exists

reflexivity : since $f'(x) = f'(x)$ $\forall x$. Then $R$ is reflexive.

symmetry : let $f \sim g \Rightarrow f'(x) = g'(x)$ $\forall x \in \mathbb{R}$

$\Rightarrow \quad g'(x) = f'(x) \Rightarrow \quad g \sim f.$

Transitivity : let $f \sim g$ & $g \sim h \Rightarrow f'(x) = g'(x)$
& $g'(x) = h'(x) \Rightarrow \forall x \in \mathbb{R} \quad f'(x) = h'(x)$
$\Rightarrow f \sim h$ Thus, $R$ is transitive.

b) $[x^2]_R = \{ g(x) \mid g'(x) = 2x \}$

$= \quad g(x) = x^2 + C, \; C \in \mathbb{R}.$

**18. (Requires calculus)**

a) Let $n$ be a positive integer. Show that the relation $R$ on the set of all polynomials with real-valued coefficients consisting of all pairs $(f, g)$ such that $f^{(n)}(x) = g^{(n)}(x)$ is an equivalence relation. [Here $f^{(n)}(x)$ is the $n$th derivative of $f(x)$.]

b) Which functions are in the same equivalence class as the function $f(x) = x^4$, where $n = 3$?

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a) reflexivity : let $f \subset P$ since $f^{(n)}(x) = f^{(n)}(x)$ $\forall x$

a) reflexivity: let $f \in P$, since $f^{(n)}(x) = f^{(n)}(x)\ \forall x$
$\Rightarrow f \sim f$ Thus, R is reflexive.

symmetry: let $f \sim g \Rightarrow f^{(n)}(x) = g^{(n)}(x)\ \forall x$
$\Rightarrow g^{(n)}(x) = f^{(n)}(x)\ \forall x \Rightarrow g \sim f$
Thus, R is symmetric

Transitivity: let $f \sim g$ & $g \sim h \Rightarrow f^{(n)}(x) = g^{(n)}(x)$
& $g^{(n)}(x) = h^{(n)}(x) \Rightarrow f^{(n)}(x) = h^{(n)}(x) \Rightarrow f \sim h$
Thus, R is transitive.

b) $[x^4]_3 = \{ g(x) \mid g^{(3)}(x) = 24x \}$

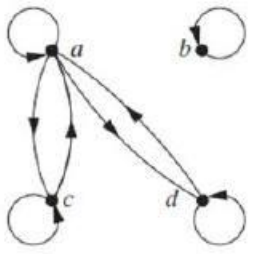
$\Rightarrow g^{(2)}(x) = \int 24x\,dx = 12x^2 + C_1$

$g^{(1)}(x) = 4x^3 + C_1 x + C_2$

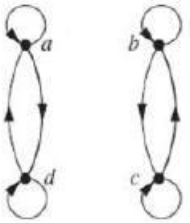
$g(x) = x^4 + \frac{C_1}{2} x^2 + C_2 x + C_3$

In Exercises **21–23** determine whether the relation with the directed graph shown is an equivalence relation.

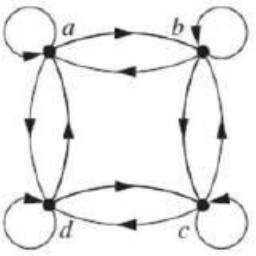
**21.**




**22.**




**23.**




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21)
- reflexive: since there are loops on each vertex
- symmetry: since there are two opposite edges between any two related vertices $\Rightarrow$ R is symmetric
- Transitivity: since triangles are not complete thus R is NOT transitive, for example

• Transitivity: Since triangles are not complete thus R is NOT transitive, for example, $(d,a), (a,c) \in R$ but $(d,c) \notin R$.

R is not an equivlance relation.

22) reflexivity: ∃ loops on each vertex ⇒ R is reflexive

symmetry: ∃ opposite direction edges between any two related vertices ⇒ R is symmetric

Transitivity: highest length of path is 2. each path of length 2 is a circuit. Thus, R is transitive.

R is a relation | classes $[a] = \{a, d\}$, $[b] = \{b, c\}$

23) reflexivity: each vertex has loops Thus, R is reflexive

symmetry: between each two vertices with relation in between ∃ two opposite arrows. Thus, R is symmetric

Transitive: ∃ path of length ② that are not supported by another path to form a triangle i.e $(d,a), (a,b) \in R$ but $(d,b) \notin R$.

**24.** Determine whether the relations represented by these zero–one matrices are equivalence relations.

a) $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

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a) • $m_{ii} = 1 \Rightarrow R$ is reflexive
• $m_{ij} \neq m_{ji}$ (i.e $m_{12} \neq m_{21}$) $\Rightarrow$ R is not symmetric

Thus, R is NOT a relation

b) • $m_{ii} = 1 \Rightarrow R$ is reflexive
• $m_{ij} \neq m_{ji}$ (i.e $m_{12} \neq m_{21}$) $\Rightarrow$ R is not symmetric

Thus, R is Not a relation

c) • $m_{ii} = 1 \Rightarrow R$ is reflexive
• $m_{ij} = m_{ji} \quad \forall i,j = 1,2,3,4 \Rightarrow R$ is symmetric
•
$$\begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \odot \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Thus, $M \odot M = M \Rightarrow M^{\odot n} = M$

Thus, $M^{\odot n} \subseteq M \quad \forall n = \{1, 2, 7, \ldots\}$

$\Rightarrow$ R is transitive

Thus, R is an equivalence relation.

**26.** What are the equivalence classes of the equivalence relations in Exercise 1?

**27.** What are the equivalence classes of the equivalence relations in Exercise 2?

**28.** What are the equivalence classes of the equivalence relations in Exercise 3?

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26) we have seen that only ⓐ and ⓒ are equivlance relations

for ⓐ The matrix was $\begin{pmatrix} 1&0&0&0 \\ 0&1&0&0 \\ 0&0&1&0 \\ 0&0&0&1 \end{pmatrix}$

Thus, each element is only related to itself

$\Rightarrow$ ∃ 4 equivdace classe

$[0], [1], [2], [3]$

for ⓒ The matrix was $\begin{pmatrix} 1&0&0&0 \\ 0&1&1&0 \\ 0&1&1&0 \\ 0&0&0&1 \end{pmatrix}$

There are three equivlece classes

we just need to look on each row. All elements with one's in a row are in the same equivlece class.

$\Rightarrow$ $[0] = \{0\}$

$[1] = \{1, 2\}$

$[3] = \{3\}$

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28) we have established that only ⓐ and ⓓ are equivlance relations.

for ⓐ $\{ (f,g) \mid f(1) = g(1) \}$

Thus, let $f : \mathbb{Z} \to \mathbb{Z}$ be an arbitrary function.

$[f]_R = \{ g(x) \mid g(1) = f(1) \}$.

for ⓑ $\{ (f,g) \mid \exists c \in \mathbb{Z}, \forall x \in \mathbb{Z}, f(x) - g(x) = c \}$

let $f: \mathbb{Z} \to \mathbb{Z}$ be an arbitrary function

$[f]_R = \{ g(x) \mid \exists c \in \mathbb{Z}, \forall x \in \mathbb{Z}, f(x) - g(x) = c \}$.

**35.** What is the congruence class $[n]_5$ (that is, the equivalence class of $n$ with respect to congruence modulo 5) when $n$ is

*a*) 2

*b*) 3

*c*) 6

*d*) $-3$

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a) These are numbers with reminder 2 with respect to 5. So, all what we need to do is add $\pm 5$ to 2.

$$[2]_5 = \{\dots, -8, -3, \underset{\downarrow}{2}, 7, 12, \dots\}$$
$$= \{2+5n \mid n \in \mathbb{Z}\}$$

b) $$[3]_5 = \{\dots, -7, -2, 3, 8, 13, 18, \dots\}$$
$$= \{3+5n \mid n \in \mathbb{Z}\}$$

c) $$[6]_5 = [1]_5 = \{\dots, -9, -4, 1, 6, 11, \dots\}$$
$$= \{1+5n \mid n \in \mathbb{Z}\}$$

d) $[-3]_5 = [2]_5$.

**36.** What is the congruence class $[n]_4$ when $n$ is

*a*) 2

*b*) 3

*c*) 6

*d*) 8

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a) $[2]_4 = \{2+4n \mid n \in \mathbb{Z}\}$

b) $[3]_4 = \{3+4n \mid n \in \mathbb{Z}\}$

c) $[6]_4 = [2]_4$

d) $[0]$ $[1]$ $[2]$ $[3]$

c) $[0]_4 = [2]_4$

d) $[8]_4 = [0]_4 = \{4n \mid n \in \mathbb{Z}\}$

**37.** Give a description of each of the congruence classes modulo 6.

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$[a]_6 = \{6n \mid n \in \mathbb{Z}\} = \{\ldots, -12, -6, 0, 6, 12, \ldots\}$

$[1]_6 = \{6n+1 \mid n \in \mathbb{Z}\}$, $[2] = \{6n+2 \mid n \in \mathbb{Z}\}$

$[m]_6 = \{6n+m \mid n \in \mathbb{Z}\}$ with $0 \leq m < 6$

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**41.** Which of these collections of subsets are partitions of $\{1, 2, 3, 4, 5, 6\}$?
*a*) $\{1, 2\}, \{2, 3, 4\}, \{4, 5, 6\}$ *b*) $\{1\}, \{2, 3, 6\}, \{4\}, \{5\}$
*c*) $\{2, 4, 6\}, \{1, 3, 5\}$ *d*) $\{1, 4, 5\}, \{2, 6\}$

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a) This is not a partion since $\{1,2\} \cap \{2,3,4\} = \{2\} \neq \phi$

b) since $\forall i,j = \{1,2,3,4\}$ $A_i \cap A_j = \phi$

$\bigcup A_i = \{1,2,3,6,4,5\} = S.$

Thus, b is a partion of S.

c) $\{2,4,6\} \cap \{1,3,5\} = \phi$

$\{2,4,6\} \cup \{1,3,5\} = \{1,2,3,4,5,6\} = S$

$\Rightarrow$ This is a partion of S.

d) $\{1,4,5\} \cap \{2,6\} = \phi$

$\{1,4,5\} \cup \{2,6\} = \{1,2,6,4,5\} \neq S$

i.e missing 3. Thus, This is not a partition of S

**42.** Which of these collections of subsets are partitions of $\{-3, -2, -1, 0, 1, 2, 3\}$?

*a*) $\{-3, -1, 1, 3\}, \{-2, 0, 2\}$

*b*) $\{-3, -2, -1, 0\}, \{0, 1, 2, 3\}$

*c*) $\{-3, 3\}, \{-2, 2\}, \{-1, 1\}, \{0\}$

*d*) $\{-3, -2, 2, 3\}, \{-1, 1\}$

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Similarly find intersections and unions.

a) $A_1 \cap A_2 = \phi$, $A_1 \cup A_2 = \{-3, -2, -1, 0, 1, 2, 3\} = S$

This is a partion

b) $A_1 \cap A_2 = \{0\} \Rightarrow$ This is not a partion.

c) $A_i \cap A_j = \phi$, $\cup A_i = \{-3, -2, -1, 0, 1, 2, 3\} = S$

$\Rightarrow$ This is a partion.

d) $A_1 \cap A_2 = \phi$ $A_1 \cup A_2 = \{-3, -2, -1, 1, 2, 3\} \neq S$

$\Rightarrow$ This is not a partion.

**46.** Which of these are partitions of the set of real numbers?

a) the negative real numbers, $\{0\}$, the positive real numbers

b) the set of irrational numbers, the set of rational numbers

c) the set of intervals $[k, k + 1]$, $k = \ldots, -2, -1, 0, 1, 2, \ldots$

d) the set of intervals $(k, k + 1)$, $k = \ldots, -2, -1, 0, 1, 2, \ldots$

e) the set of intervals $(k, k + 1]$, $k = \ldots, -2, -1, 0, 1, 2, \ldots$

f ) the sets $\{x + n \mid n \in Z\}$ for all $x \in [0, 1)$

a) True $A_i \cap A_j = \emptyset \quad \forall i \neq j$

$\bigcup A_i = \mathbb{R}$

b) True $\mathbb{R} = Q \cup Q' \qquad Q \cap Q' = \emptyset$

c) $k \in \mathbb{Z}$ consider two intervals

$[k, k+1] \qquad [k+1, k+2]$

Since $[k, k+1] \cap [k+1, k+2] = \{k\} \neq \emptyset$

and $[k, k+1]$, $[k+1, k+2]$ are two intervals

$\Rightarrow$ This is not a partion.

d) $\forall k \in \mathbb{Z} \qquad (k, k+1) \cap (k+1, k+2) = \emptyset$

$\bigcup_k A_k = \bigcup_k (k, k+1) = \mathbb{R} - \mathbb{Z}$

Since we add all numbers in the interval but miss the boundry point $\{k, k+1\}$. $\forall$ intervals.

e) notice $\forall k \in \mathbb{Z} \qquad (k, k+1] \cap (k+1, k+2] = \emptyset$
furthermore, $\bigcup_k (k, k+1] = \mathbb{R}$.

intervals produce points between two integers and the integers are formed by the boundry point $k+1$.

f) $\{x+n \mid n \in \mathbb{Z} \ \forall, x \in [0, 1)\}$

we can write these sets as $[n, n+1)$

Notice $[n, n+1) \cap [n+1, n+2) = \emptyset \quad \forall n$
Thus, all these subsets are disjoint

also, $\bigcup_n [n, n+1) = \mathbb{R}$
Since each interval produce the part between $n$ and $n+1$

also, $\bigcup_n [n, n+1] = \mathbb{R}$

Since each interval produce the part between n and n+1 and the set of integs are formed from one boundary n.

**47.** List the ordered pairs in the equivalence relations produced by these partitions of $\{0, 1, 2, 3, 4, 5\}$.

*a*) $\{0\}, \{1, 2\}, \{3, 4, 5\}$

*b*) $\{0, 1\}, \{2, 3\}, \{4, 5\}$

*c*) $\{0, 1, 2\}, \{3, 4, 5\}$

*d*) $\{0\}, \{1\}, \{2\}, \{3\}, \{4\}, \{5\}$

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a) we just need to take every subset and produce pairs that are reflexive, symmetric and transitive.

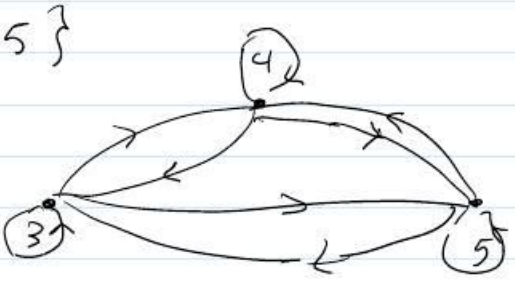
1) $\{0\} \Rightarrow \{(0,0)\} = A_1$

2) $\{1,2\} \Rightarrow \{(1,1), (2,2)$ ← reflexive
$(1,2), (2,1)\}$ ← symetric $= A_2$
← transitive

one other way is to produce complete directed graphs with loops for the points




3) $\{3,4,5\}$



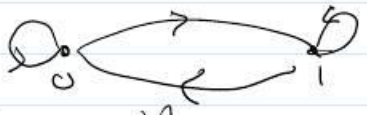

$\{(3,3), (4,4), (5,5), (3,4), (4,3), (4,5)$
$(5,4), (3,5), (5,3)\} = A_3$

$\Rightarrow A = A_1 \cup A_2 \cup A_3$

b) $\{0,1\}$



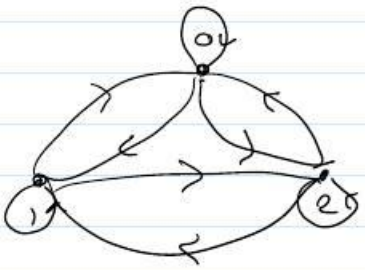
b) $\{0,1\}$




$\{(0,1), (1,1), (1,0), (0,1)\}$

similarly for all other sets and the set is the union of all these sets.

c) $\{0,1,2\}$




$\{(0,0), (1,1), (2,2), (1,0), (0,1), (0,2), (2,0), (1,2), (2,1)\}$

similarly for the other class.

d) $[1] = \{(1,1)\}$ similarly for all other numbers.

**48.** List the ordered pairs in the equivalence relations produced by these partitions of $\{a, b, c, d, e, f, g\}$.

*a*) $\{a, b\}, \{c, d\}, \{e, f, g\}$

*b*) $\{a\}, \{b\}, \{c, d\}, \{e, f\}, \{g\}$

*c*) $\{a, b, c, d\}, \{e, f, g\}$

*d*) $\{a, c, e, g\}, \{b, d\}, \{f\}$

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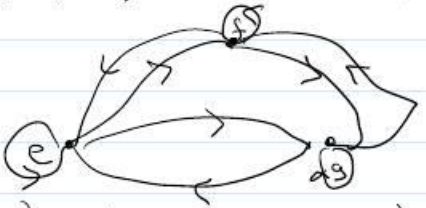
a) 1) $\{a,b\}$




$\{(a,b), (b,b), (a,b), (b,a)\} = A_1$

similarly for $\{c,d\} \Rightarrow \{(c,c), (d,d), (c,d), (d,c)\} = A_2$

2) $\{e,f,g\} \Rightarrow$




$\{(e,e), (f,f), (g,g), (e,f), (f,e), (e,g), (g,e)$



$\{(e,e), (f,f), (g,g), (e,f), (f,e), (e,g), (g,e), (f,g), (g,f)\} = A_3$

$\Rightarrow \Delta = A_1 \cup A_2 \cup A_3$

b) $\{a\}$, $\{b\}$, $\{g\}$ $\longrightarrow$ 

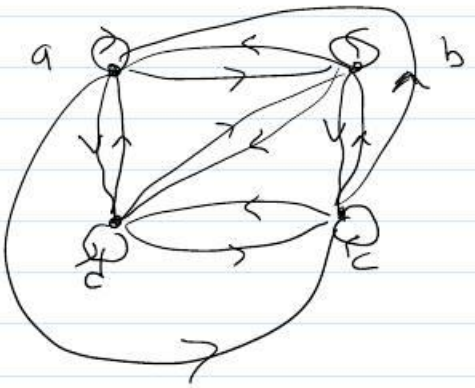
Thus, They produce $\{(a,a), (b,b), (g,g)\}$

$\{c,d\}, \{e,f\} \longrightarrow$ 

Thus, $\{(c,c), (d,d), (e,e), (f,f), (c,d), (d,c), (e,f), (f,e)\}$

Total set = $\{(a,a), (b,b), (g,g), (c,c), (d,d), (e,e), (f,f), (c,d), (d,c), (e,f), (f,e)\}$.

c) $\{a,b,c,d\}$




$\Rightarrow \{(a,a), (b,b), (c,c), (d,d), (a,b), (a,c), (a,d), (b,a), (b,c), (b,d), (d,a), (d,b), (d,c), (c,a), (c,b), (c,d)\}$

$\{e,f,g\} \Rightarrow \{(e,e), (e,f), (e,g), (f,f), (f,g), (g,g), (g,f), (g,e), (f,e)\}$

d) Similar method you can use any of the two discussed methods.